Universiteit Leiden, LIACS

Meeting 1

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Date:	11 Mar 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
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Outcome of previous meeting

Previous meeting, we talked about the idea of the thesis proposal, as well as how to approach the research questions. From this meeting came forth:

- To sort out the bureaucracy (including the contract, and an elaborate proposal with more specifications of a planning)
- To extent the number of baseline frameworks (beyond the three given in the original proposal)
- Results of these frameworks on a single algorithm (PPO).
- A list of environments suitable for testing consistency across baseline frameworks.

Results during the last weeks

- The bureaucracy: contract proposal is signed, and planning is updated with feedback received via teams
- The number of frameworks has been extended, not just by arbitrary choice, but now looking more into related work (and other specifications giving reasoning as to continue with experimenting with this framework or not). This yields a list of properties of the frameworks, along with a list of algorithms each of these frameworks has implemented.
- Results of these frameworks on PPO, benchmarked and reported by each framework (if applicable), via the paper, GitHub, or documentation, along with (known) experimental parameters. These results can be sorted per environment and compared, for easy outlier detection.
- Libraries for experimenting are known, specific environments are yet to be determined.

Plans for next weeks

- Look into the comparability of results of different versions of the same environment
- Finalize a list of frameworks, such that more focussed research can be done in its motivation and point out specific implementation choices.
- Finalize a list of environments, based on popularity of its use among frameworks, and (unexplained variable) inconsistencies among results of these environments. These environments should be diverse enough as to be too easily solved (such as Pong), and performance directly related to the algorithm (and not other factors in environments such as Montezuma's revenge). Try to find a reasoning for the outliers present in the data. Environment version control is part of this.
- Set up a plan to run a single iteration of PPO on a single environment of each framework.